

Prathyum G (U25U23T040073)

1. choose a suitable load cell: **Strain Gauge Load Cell (Single-Point, 4-wire type), 5kg** with the following specification:-

- **Load Capacity** : let us choose **1.5 times** the maximum expected weight (food + pan weight) which is **5kg**
- **Accuracy**: **±0.05% FSO** Class C3 load cell
- **Sensitivity**: **1.5 ±0.15** mV/V gives a better sensitivity
- **Noise / Disturbance Resistance**: **IP65/IP67**

2. ADC : **ADS1231** with specifications :-

- **Resolution** : **24-bit**
- **InterfaceSimple**: **2-wire**
- **Sample Rate** : **10 / 40 SPS**
- **Brand** : **Texas Instruments**
- **Cost** : **\$1-2**

3. Connectors And Cables

- **Load Cell Cable** :

Shielded Twisted Pair (STP), 24 AWG, 4+Shield

Minimizes electromagnetic interference and ensures accurate low-level signal transmission core

- **Load Cell to ADS1231** :

GX16 Aviation Connector

- **ADS1231 to MCU** :

JST-XH series, 2.54mm,3A,250V 4-pin(VCC, GND,SCLK,DATA)

Compact, reliable, suitable for PCB-based modular connections

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4. CODE

```
#define SCLK 3 // Clock pin
#define DATA 2 // Data pin
// Calibration
long zeroOffset = 0; // For tare function
float calibFactor = 420.0; // Adjust after physical calibration
void setup()
{
  Serial.begin(9600);
  pinMode(SCLK, OUTPUT);
  pinMode(DATA, INPUT);
  // Take zero reading at startup (tare)
  zeroOffset = readADS1231();
  Serial.println("Weight Measurement Started!");
  Serial.println(" ");
}
// Read 24-bit value from ADS1231
long readADS1231()
{
  long result = 0;
  while(digitalRead(DATA) == HIGH); // Wait for data
  // Read 24 bits one by one
  for(int i = 0; i < 24; i++) {
    digitalWrite(SCLK, HIGH);
    result = (result << 1) | digitalRead(DATA);
  }
}
```

Prathyum G (U25U23T040073)

```
digitalWrite(SCLK, LOW);
}
// Extra pulse to finish
digitalWrite(SCLK, HIGH);
digitalWrite(SCLK, LOW);
return result;
}

void loop() {
    long raw = readADS1231();
    float weight = (raw - zeroOffset) / calibFactor;    // Subtract zero offset (tare)
    if(weight < 0) weight = 0.0;
    Serial.print("Raw Value : ");
    Serial.println(raw);
    Serial.print("Weight   : ");
    Serial.print(weight, 3);                          // 3 decimal places
    Serial.println(" kg");
    Serial.println(" ");
    delay(1000);                                        // Read every 1 second
}
```

5. PCB designing files including schematics and PCB board files are uploaded BELOW

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